Martingales — Solutions (set 3)

Problem I — Fixation Probability and Accumulated Genetic Variance in a Boundary-Slowed Genetic Drift Model (19 pts)

Preliminaries. Since 0 and N are absorbing and $X_0 = x$ is deterministic, we have $X_n \in E = \{0, \dots, N\}$ for all n, hence $0 \le X_n \le N$: the chain is uniformly bounded. For interior k, the increment $X_{n+1} - X_n$ equals $+1$, -1 or 0, with

$$\mathbb{E}[X_{n+1} - X_n \mid X_n = k] = a_k \cdot (+1) + a_k \cdot (-1) + (1 - 2a_k) \cdot 0 = 0,$$

$$\mathbb{E}[(X_{n+1} - X_n)^2 \mid X_n = k] = a_k \cdot 1 + a_k \cdot 1 + (1 - 2a_k) \cdot 0 = 2a_k = v_k.$$

These two identities, which also hold trivially at $k \in \{0, N\}$ (where the increment is zero and $v_0 = v_N = 0$), are the foundation of everything that follows. By the Markov property, for every bounded function $g : E \to \mathbb{R}$,

$$\mathbb{E}[g(X_{n+1}) \mid \mathcal{F}_n] = \mathbb{E}[g(X_{n+1}) \mid X_n] \qquad \text{a.s.}$$

(the law of X_{n+1} depends on the past only through X_n).

Question 1.

(a) Stopping time and finiteness. We have $\{\tau = n\} = \{X_0 \notin \{0, N\}, \dots, X_{n-1} \notin \{0, N\},\ X_n \in \{0, N\}\}$, which is expressed in terms of $X_0, \dots, X_n$, hence belongs to $\mathcal{F}_n$. Thus τ is a stopping time.

Let us show $\mathbb{P}(\tau < \infty) = 1$. Set $c := \big(\min_{1 \le j \le N-1} a_j\big)^N > 0$ (the minimum of a finite number of quantities $a_j = 2j(N-j)/N^2 > 0$) and $m := N$. From any interior state k, by stepping $+1$ at each step while interior, one reaches the boundary N in at most $N - 1 \le m$ steps; now at each step in an interior state j the probability of stepping up is $a_j \ge \min_j a_j$, so the probability of this upward trajectory is $\ge \big(\min_j a_j\big)^{N-1} \ge c$. Thus $\mathbb{P}_k(\tau \le m) \ge c$ for every interior state k, and trivially $\mathbb{P}_k(\tau \le m) = 1 \ge c$ for $k \in \{0, N\}$. By the Markov property applied at time nm,

$$\mathbb{P}(\tau > (n+1)m \mid \mathcal{F}_{nm}) = \mathbf{1}_{\{\tau > nm\}}\, \mathbb{P}_{X_{nm}}(\tau > m) \le (1 - c)\, \mathbf{1}_{\{\tau > nm\}} \quad \text{a.s.},$$

because on $\{\tau > nm\}$ the state X_{nm} is interior. Taking expectations, $\mathbb{P}(\tau > (n+1)m) \le (1-c)\, \mathbb{P}(\tau > nm)$, whence by induction $\mathbb{P}(\tau > nm) \le (1 - c)^n \to 0$. Therefore $\mathbb{P}(\tau = \infty) = \lim_n \mathbb{P}(\tau > nm) = 0$, i.e. $\mathbb{P}(\tau < \infty) = 1$. (The same geometric argument moreover gives $\mathbb{E}[\tau] < \infty$, a fact we shall reuse.)

(b) $M_n = X_n$ is a bounded martingale. The process (X_n) is adapted (by definition of $\mathcal{F}_n$) and bounded, hence integrable: $|X_n| \le N$ and $\mathbb{E}|X_n| \le N < \infty$. Let us verify the martingale property. By the Markov property and then the preliminary computation with $g = \mathrm{id}$,

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[X_{n+1} \mid X_n].$$

On $\{X_n = k\}$ with k interior, this equals $k + \mathbb{E}[X_{n+1} - X_n \mid X_n = k] = k = X_n$. On $\{X_n \in \{0, N\}\}$, the state is absorbing so $X_{n+1} = X_n$ and the equality is immediate. In all cases

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n \quad \text{a.s.},$$

which establishes that $(M_n) = (X_n)$ is a martingale. It is bounded by N. □

Remark. The function $h(k) = k$ is *harmonic* for the chain (it satisfies the harmonicity condition at every interior k), and it is precisely this harmonicity that makes $h(X_n)$ a martingale. The choice

$a_k = 2k(N-k)/N^2$ makes the chain *symmetric* (equal probabilities of stepping up and down), whence the martingale character, but with an increment variance $v_k = 2a_k$ depending on the state: the chain "slows down" near the boundaries.

Question 2. The time τ is a stopping time, finite a.s. (Question 1(a)), and (M_n) is a bounded martingale. We apply the optional stopping theorem in its form valid for bounded martingales and a.s. finite stopping times. Precisely, the stopped process $(M_{n\wedge\tau})$ is a martingale, so $\mathbb{E}[M_{n\wedge\tau}] = \mathbb{E}[M_0] = x$ for all n. Since $\tau < \infty$ a.s., we have $M_{n\wedge\tau} \to M_\tau = X_\tau$ a.s. as $n \to \infty$; and $|M_{n\wedge\tau}| \le N$ uniformly, so by dominated convergence

$$\mathbb{E}[X_\tau] = \lim_{n\to\infty} \mathbb{E}[M_{n\wedge\tau}] = x.$$

Now $X_\tau \in \{0, N\}$ (by definition of τ and a.s. finiteness of τ), so $X_\tau = N \cdot \mathbf{1}_{\{X_\tau = N\}}$ and

$$\mathbb{E}[X_\tau] = N\,\mathbb{P}(X_\tau = N) = N\,u(x).$$

Combining, $N\,u(x) = x$, whence

$$u(x) = \mathbb{P}(X_\tau = N) = \frac{x}{N}\,.$$

The fixation probability is thus simply proportional to the initial frequency, exactly as for the symmetric random walk: the slowing at the boundaries does not affect the absorption probabilities (only the times). $\square$

Question 3.

(a) (Y_n) is a martingale. The process $Y_n = X_n^2 - A_n$ is adapted: X_n is, and $A_n = \sum_{i=0}^{n-1} v_{X_i}$ is $\mathcal{F}_{n-1}$-measurable, hence $\mathcal{F}_n$-measurable. It is bounded, hence integrable: $0 \le X_n^2 \le N^2$, and $0 \le A_n = \sum_{i=0}^{n-1} v_{X_i} \le n \max_k v_k \le n$ (since $v_k = 4k(N-k)/N^2 \le 1$, the maximum of $k(N-k)$ being $\le N^2/4$), whence $\mathbb{E}|Y_n| < \infty$.

Let us compute the conditional increment. Since $A_{n+1} - A_n = v_{X_n}$ is $\mathcal{F}_n$-measurable,

$$\mathbb{E}[Y_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[X_{n+1}^2 \mid \mathcal{F}_n] - A_n - v_{X_n}.$$

By the Markov property, $\mathbb{E}[X_{n+1}^2 \mid \mathcal{F}_n] = \mathbb{E}[X_{n+1}^2 \mid X_n]$. Write $X_{n+1}^2 = X_n^2 + 2X_n(X_{n+1} - X_n) + (X_{n+1} - X_n)^2$. Conditioning on $X_n = k$ and using the two preliminary identities ($\mathbb{E}[X_{n+1} - X_n \mid X_n = k] = 0$ and $\mathbb{E}[(X_{n+1} - X_n)^2 \mid X_n = k] = v_k$, valid including at the boundaries):

$$\mathbb{E}[X_{n+1}^2 \mid X_n = k] = k^2 + 2k \cdot 0 + v_k = k^2 + v_k.$$

Hence $\mathbb{E}[X_{n+1}^2 \mid \mathcal{F}_n] = X_n^2 + v_{X_n}$ a.s., and consequently

$$\mathbb{E}[Y_{n+1} \mid \mathcal{F}_n] = \left(X_n^2 + v_{X_n}\right) - A_n - v_{X_n} = X_n^2 - A_n = Y_n \quad \text{a.s.}$$

Thus (Y_n) is a martingale. (In other words, A_n is the predictable *compensator* — in the sense of the Doob decomposition — of the submartingale X_n^2.) $\square$

(b) Computation of $\mathbb{E}[A_\tau]$. The stopped process $(Y_{n\wedge\tau})$ is a martingale, so for all n,

$$\mathbb{E}[Y_{n\wedge\tau}] = \mathbb{E}[Y_0] = X_0^2 - A_0 = x^2.$$

Rewriting: $\mathbb{E}[X_{n\wedge\tau}^2] - \mathbb{E}[A_{n\wedge\tau}] = x^2$, that is

$$\mathbb{E}[A_{n\wedge\tau}] = \mathbb{E}[X_{n\wedge\tau}^2] - x^2.$$

Let us pass to the limit $n \to \infty$. Since $\tau < \infty$ a.s., $X_{n\wedge\tau} \to X_\tau$ a.s., and $|X_{n\wedge\tau}^2| \le N^2$; by dominated convergence,

$$\mathbb{E}[X_{n\wedge\tau}^2] \xrightarrow[n\to\infty]{} \mathbb{E}[X_\tau^2].$$

Now $X_\tau \in \{0, N\}$ with $\mathbb{P}(X_\tau = N) = x/N$ (Question 2), so

$$\mathbb{E}[X_\tau^2] = N^2 \cdot \frac{x}{N} = Nx.$$

For the left-hand side, $A_{n\wedge\tau}$ is increasing in n (the $v_{X_i} \geq 0$) and $A_{n\wedge\tau} \uparrow A_\tau$ a.s. By monotone convergence,

$$\mathbb{E}[A_{n\wedge\tau}] \xrightarrow[n\to\infty]{} \mathbb{E}[A_\tau].$$

(This limit is finite: $A_\tau = \sum_{i=0}^{\tau-1} v_{X_i} \leq \tau$ and $\mathbb{E}[\tau] < \infty$, which confirms the integrability of A_τ.) Passing to the limit in the identity,

$$\mathbb{E}[A_\tau] = \mathbb{E}[X_\tau^2] - x^2 = Nx - x^2,$$

so finally

$$\mathbb{E}[A_\tau] = \mathbb{E}\Big[\sum_{i=0}^{\tau-1} v_{X_i}\Big] = x\,(N-x)\,.$$

Interpretation. The total accumulated genetic variance before fixation equals exactly $x(N-x)$, maximal when the population starts from a balanced frequency $x = N/2$ (value $N^2/4$) and zero at the boundaries, which is consistent with the idea that drift "works" the most when the two alleles coexist in equal proportions. ■

Problem II — The Ballot Problem via a Reversed Martingale (20 pts)

Preliminaries. The counting order is a uniform permutation of the $n = a + b$ ballots (a for A, b for B). The sequence $(\xi_1, \dots, \xi_n)$ is therefore *exchangeable*: its law is invariant under permutation of the indices, and exactly a of the ξ_i equal $+1$. We have $L_k = \sum_{i=1}^k \xi_i$, $L_0 = 0$, $L_n = a - b$.

Since $\sum_{i=1}^n \xi_i = a - b$ is *constant*, we have, for all k,

$$L_k = \sum_{i=1}^k \xi_i = (a-b) - \sum_{i=k+1}^n \xi_i,$$

which is a function of $(\xi_{k+1}, \dots, \xi_n)$: *hence L_k is $\mathcal{G}_k$-measurable.* In particular $M_k = L_k/k$ is $\mathcal{G}_k$-measurable, and $|M_k| = |L_k|/k \le 1$, so M_k is bounded, hence integrable.

Question (1). The measurability of L_k has just been established. Fix $k \in \{1, \dots, n\}$. Let A_k, B_k denote the numbers of A and B votes among the first k ballots. Then $A_k + B_k = k$ and $A_k - B_k = L_k$, whence

$$A_k = \frac{k + L_k}{2}, \qquad B_k = \frac{k - L_k}{2}.$$

Conditionally on $\mathcal{G}_k = \sigma(\xi_{k+1}, \dots, \xi_n)$, the composition of the "tail" (positions $k+1, \dots, n$) is known; by complementarity, that of the first k ballots is also known, so A_k and B_k are $\mathcal{G}_k$-measurable. By exchangeability, conditionally on the composition of the tail, the block $(\xi_1, \dots, \xi_k)$ is a *uniform draw* among all arrangements of A_k symbols $+1$ and B_k symbols -1. For such a uniform draw without replacement, the law of *any* coordinate, in particular of the k-th, is: $\mathbb{P}(\xi_k = +1 \mid \mathcal{G}_k) = A_k/k$. Hence

$$\mathbb{E}\big[\mathbf{1}_{\{\xi_k = +1\}} \mid \mathcal{G}_k\big] = \frac{A_k}{k} = \frac{k + L_k}{2k}.$$

Question (2). Let $k \in \{1, \dots, n-1\}$. We apply the formula of (1) to the index $k+1$, with $\mathcal{G}_{k+1} = \sigma(\xi_{k+2}, \dots, \xi_n)$:

$$\mathbb{E}\big[\mathbf{1}_{\{\xi_{k+1} = +1\}} \mid \mathcal{G}_{k+1}\big] = \frac{(k+1) + L_{k+1}}{2(k+1)}, \quad \mathbb{E}\big[\mathbf{1}_{\{\xi_{k+1} = -1\}} \mid \mathcal{G}_{k+1}\big] = \frac{(k+1) - L_{k+1}}{2(k+1)},$$

whence, since $\xi_{k+1} = \mathbf{1}_{\{\xi_{k+1} = +1\}} - \mathbf{1}_{\{\xi_{k+1} = -1\}}$,

$$\mathbb{E}\big[\xi_{k+1} \mid \mathcal{G}_{k+1}\big] = \frac{L_{k+1}}{k+1}.$$

Since $L_k = L_{k+1} - \xi_{k+1}$ and L_{k+1} is $\mathcal{G}_{k+1}$-measurable,

$$\mathbb{E}\big[L_k \mid \mathcal{G}_{k+1}\big] = L_{k+1} - \frac{L_{k+1}}{k+1} = L_{k+1} \cdot \frac{k}{k+1}.$$

Dividing by k:

$$\mathbb{E}\big[M_k \mid \mathcal{G}_{k+1}\big] = \frac{1}{k}\,\mathbb{E}\big[L_k \mid \mathcal{G}_{k+1}\big] = \frac{L_{k+1}}{k+1} = M_{k+1}.$$

Each M_k being $\mathcal{G}_k$-measurable and bounded by 1, the process $(M_k)_{1 \le k \le n}$ is indeed a martingale with respect to the decreasing filtration $(\mathcal{G}_k)$: a *reversed martingale.*

"Reversed time" viewpoint. Set $s = n - k$ and $\widetilde{M}_s = M_{n-s}$, $\widetilde{\mathcal{F}}_s = \mathcal{G}_{n-s}$ for $s \in \{0, \dots, n-1\}$. Then $(\widetilde{\mathcal{F}}_s)$ is an *increasing* filtration and the identity $\mathbb{E}[M_k \mid \mathcal{G}_{k+1}] = M_{k+1}$ reads $\mathbb{E}[\widetilde{M}_{s+1} \mid \widetilde{\mathcal{F}}_s] = \widetilde{M}_s$: $(\widetilde{M}_s)_{0 \le s \le n-1}$ is an *ordinary* martingale, with initial value $\widetilde{M}_0 = M_n$.

Question (3). For $k \in \{1, \dots, n\}$, by definition of R (last visit to 0, conv. $R = 1$ otherwise),

$$\{R \le k\} = \{\, L_j \neq 0 \ \text{ for all } j \in \{k+1, \dots, n\}\,\}.$$

Indeed: if there exists a zero among $\{1, \dots, n\}$, then $R \le k$ means that no zero occurs after time k; and if none exists (event $E = \{L_j > 0\ \forall j\}$, see Q4), then $R = 1 \le k$ and the right-hand side also holds (no L_j is zero). Now, for $j \ge k+1$, we have $L_j = (a-b) - \sum_{i=j+1}^{n} \xi_i$, which involves only $\xi_{j+1}, \dots, \xi_n$ with $j+1 \ge k+2$; each such L_j is therefore $\mathcal{G}_k = \sigma(\xi_{k+1}, \dots, \xi_n)$-measurable. Thus

$$\{R \le k\} \in \mathcal{G}_k \qquad \text{for all } k \in \{1, \dots, n\},$$

that is, R is a stopping time for the reversed filtration. (Equivalently, in reversed time $s = n-k$, the time $\sigma = n - R \in \{0, \dots, n-1\}$ satisfies $\{\sigma \le s\} = \{R \ge n-s\} \in \widetilde{\mathcal{F}}_s$, so σ is a bounded stopping time for the increasing filtration $(\widetilde{\mathcal{F}}_s)$.)

Optional stopping theorem applied. The process $(\widetilde{M}_s)_{0 \le s \le n-1}$ is a bounded (ordinary) martingale, and $\sigma = n - R$ is a bounded stopping time ($0 \le \sigma \le n-1$). Doob's optional stopping theorem, in its version for *bounded stopping times* (or for a uniformly integrable martingale, which holds automatically here since $|\widetilde{M}_s| \le 1$), gives

$$\mathbb{E}[\widetilde{M}_\sigma] = \mathbb{E}[\widetilde{M}_0], \qquad \text{i.e.} \qquad \mathbb{E}[M_R] = \mathbb{E}[M_n].$$

The hypotheses are indeed satisfied: a martingale bounded by 1 and a stopping time bounded by $n-1$.

Question (4). We compute M_R according to the two complementary events.

(i) On $E = \{L_k > 0\ \forall k \in \{1, \dots, n\}\}$ (A always strictly in the lead), the set $\{k \in \{1, \dots, n\} : L_k = 0\}$ is empty, so $R = 1$ by convention. Moreover $L_1 > 0$ and $L_1 \in \{\pm 1\}$ force $L_1 = +1$, whence

$$M_R = M_1 = \frac{L_1}{1} = 1 \qquad \text{on } E.$$

(ii) On E^c, there exists $k \in \{1, \dots, n\}$ with $L_k \le 0$. Since $L_n = a - b > 0$ and the steps equal ± 1, in passing from a value ≤ 0 to the final value > 0 the walk necessarily takes the value 0 at some time in $\{1, \dots, n\}$ (the first k such that $L_k \le 0$ satisfies $L_k = 0$, since $L_{k-1} \ge 1$ and the step equals ± 1). Hence the set defining R is nonempty, $R \ge 2$ is its maximum, and $L_R = 0$; thus

$$M_R = \frac{L_R}{R} = 0 \qquad \text{on } E^c.$$

Combining, $M_R = \mathbf{1}_E \cdot 1 + \mathbf{1}_{E^c} \cdot 0 = \mathbf{1}_E$ almost surely, whence

$$\mathbb{E}[M_R] = \mathbb{P}(E) = p.$$

On the other hand $M_n = L_n/n = (a-b)/(a+b)$ is deterministic, so $\mathbb{E}[M_n] = (a-b)/(a+b)$. The equality $\mathbb{E}[M_R] = \mathbb{E}[M_n]$ from question (3) then gives

$$\boxed{p = \frac{a-b}{a+b}}.$$

Remark (check). This is Bertrand's *ballot theorem.* Let us check $a = 2, b = 1$ ($n = 3$): the $\binom{3}{1} = 3$ equally likely orders are AAB, ABA, BAA. Only AAB gives $L_1{=}1, L_2{=}2, L_3{=}1$ all > 0; for ABA we have $L_2 = 0$, for BAA we have $L_1 = -1$. Hence $p = 1/3 = (2-1)/(2+1)$. Consistent.

Problem III — Empty boxes, Doob martingale and Azuma–Hoeffding concentration (21 pts)

Question 1. For each box j, set $A_j = \{X_i \neq j \ \forall i\} = \{\text{box } j \text{ is empty}\}$, so that $Z = \sum_{j=1}^n \mathbf{1}_{A_j}$. Since the X_i are i.i.d. uniform on $\{1, \dots, n\}$,

$$\mathbb{P}(A_j) = \prod_{i=1}^{n} \mathbb{P}(X_i \neq j) = \left(1 - \tfrac{1}{n}\right)^n,$$

which does not depend on j. By linearity of expectation,

$$\mathbb{E}[Z] = \sum_{j=1}^{n} \mathbb{P}(A_j) = n\left(1 - \tfrac{1}{n}\right)^n.$$

Finally $\mathbb{E}[Z]/n = (1 - 1/n)^n = \exp\bigl(n \ln(1 - 1/n)\bigr)$, and $n \ln(1 - 1/n) = n\bigl(-\frac{1}{n} + O(n^{-2})\bigr) \to -1$, whence $\mathbb{E}[Z]/n \to e^{-1}$.

Question 2.

(a) Z is a bounded function of the X_i ($0 \leq Z \leq n$), hence integrable, and $M_k = \mathbb{E}[Z \mid \mathcal{F}_k]$ is well defined and $\mathcal{F}_k$-measurable. It is the Doob martingale of Z: for $1 \leq k \leq n$, by the tower property ($\mathcal{F}_{k-1} \subset \mathcal{F}_k$),

$$\mathbb{E}[M_k \mid \mathcal{F}_{k-1}] = \mathbb{E}\bigl[\mathbb{E}[Z \mid \mathcal{F}_k] \mid \mathcal{F}_{k-1}\bigr] = \mathbb{E}[Z \mid \mathcal{F}_{k-1}] = M_{k-1}.$$

Moreover $\mathcal{F}_0$ is trivial, so $M_0 = \mathbb{E}[Z]$, and since Z is $\mathcal{F}_n$-measurable, $M_n = \mathbb{E}[Z \mid \mathcal{F}_n] = Z$.

(b) Write $Z = f(X_1, \dots, X_n)$ with

$$f(x_1, \dots, x_n) = \sum_{j=1}^{n} \mathbf{1}\{x_i \neq j \ \forall i\}.$$

Bounded difference property. Fix an index k and two n-tuples $x = (x_1, \dots, x_n)$ and $x' = (x_1, \dots, x_{k-1}, x'_k, x_{k+1}, \dots, x_n)$ that differ only in the k-th coordinate. In passing from x to x', a single ball is moved: box x_k may become empty (gain ≤ 1) and box x'_k may cease to be empty (loss ≤ 1), the other boxes being unchanged. Hence

$$|f(x) - f(x')| \leq 1.$$

Passage to the Doob martingale. Let X'_k be a copy of X_k independent of $(X_1, \dots, X_n)$, and set

$$Z' = f(X_1, \dots, X_{k-1}, X'_k, X_{k+1}, \dots, X_n).$$

Since Z' does not involve X_k, and $(X'_k, X_{k+1}, \dots, X_n)$ is independent of X_k conditionally on $\mathcal{F}_{k-1}$,

$$\mathbb{E}[Z' \mid \mathcal{F}_k] = \mathbb{E}[Z' \mid \mathcal{F}_{k-1}].$$

Now X'_k has the same law as X_k and is, like X_k, independent of $\mathcal{F}_{k-1}$; hence $(X_1, \dots, X_{k-1}, X'_k, X_{k+1}, \dots, X_n)$ has the same conditional law, given $\mathcal{F}_{k-1}$, as $(X_1, \dots, X_n)$. Consequently

$$\mathbb{E}[Z' \mid \mathcal{F}_{k-1}] = \mathbb{E}[Z \mid \mathcal{F}_{k-1}] = M_{k-1}, \quad \text{whence} \quad \mathbb{E}[Z' \mid \mathcal{F}_k] = M_{k-1}.$$

Thus, using $M_k = \mathbb{E}[Z \mid \mathcal{F}_k]$,

$$M_k - M_{k-1} = \mathbb{E}[Z \mid \mathcal{F}_k] - \mathbb{E}[Z' \mid \mathcal{F}_k] = \mathbb{E}[\, Z - Z' \mid \mathcal{F}_k \,].$$

Now $|Z - Z'| \le 1$ a.s. by the bounded difference property (the vectors defining Z and Z' differ only in coordinate k). The conditional Jensen inequality (or simply the monotonicity of conditional expectation) then gives

$$|M_k - M_{k-1}| = \big|\mathbb{E}[Z - Z' \mid \mathcal{F}_k]\big| \le \mathbb{E}\big[\, |Z - Z'| \;\big|\; \mathcal{F}_k\big] \le 1.$$

Question 3.

(a) We apply Azuma–Hoeffding to the martingale $(M_k)_{0 \le k \le n}$ of Question 2, with $c_k = 1$ for all k, so $\sum_{k=1}^n c_k^2 = n$. Since $M_n - M_0 = Z - \mathbb{E}[Z]$, we obtain for all $t \ge 0$

$$\mathbb{P}\big(Z - \mathbb{E}[Z] \ge t\big) \le e^{-t^2/(2n)}.$$

Applying the same result to the martingale $(-M_k)$ (which satisfies the same bounds $|-M_k - (-M_{k-1})| \le 1$), we also have $\mathbb{P}\big(Z - \mathbb{E}[Z] \le -t\big) \le e^{-t^2/(2n)}$. By the union of the two events,

$$\mathbb{P}\big(|Z - \mathbb{E}[Z]| \ge t\big) \le 2\, e^{-t^2/(2n)}.$$

(b) Fix $\varepsilon > 0$. By Question 1, $a_n := \mathbb{E}[Z]/n = (1 - 1/n)^n \to e^{-1}$; hence there exists n_0 such that for $n \ge n_0$ we have $|a_n - e^{-1}| \le \varepsilon/2$. For such an n,

$$\Big\{\big|\tfrac{Z}{n} - e^{-1}\big| > \varepsilon\Big\} \subset \Big\{\big|\tfrac{Z}{n} - a_n\big| > \tfrac{\varepsilon}{2}\Big\} = \Big\{|Z - \mathbb{E}[Z]| > \tfrac{\varepsilon n}{2}\Big\}.$$

We apply (a) with $t = \varepsilon n/2$:

$$\mathbb{P}\Big(\big|\tfrac{Z}{n} - e^{-1}\big| > \varepsilon\Big) \le 2\exp\!\Big(-\frac{(\varepsilon n/2)^2}{2n}\Big) = 2\exp\!\Big(-\frac{\varepsilon^2 n}{8}\Big) \xrightarrow[n\to\infty]{} 0.$$

The convergence to 0 is *exponential in* n (at rate $\varepsilon^2/8$), which in particular establishes $Z/n \to e^{-1}$ in probability. (The summability of these bounds over n together with Borel–Cantelli would even yield almost sure convergence.)

Problem IV — Poisson Galton–Watson process: extinction probability and the martingale $W_n = Z_n/\lambda^n$ (19 pts)

Preliminaries. The branching structure gives, conditionally on $\mathcal{F}_n$, that Z_{n+1} is a sum of Z_n i.i.d. copies of $\xi \sim \text{Poisson}(\lambda)$, independent of $\mathcal{F}_n$. We will constantly use the following identity: if N is $\mathcal{F}_n$-measurable with values in $\mathbb{N}$ and $(\xi_i)_{i\geq 1}$ are i.i.d., independent of $\mathcal{F}_n$, then for the conditional generating function

$$\mathbb{E}\Big[s^{\sum_{i=1}^N \xi_i} \,\Big|\, \mathcal{F}_n\Big] = \big(f(s)\big)^N \qquad \text{a.s.,} \qquad f(s) = \mathbb{E}[s^\xi].$$

Question 1.

(a) Computation of f. By definition,

$$f(s) = \sum_{\ell\geq 0} s^\ell \, e^{-\lambda}\frac{\lambda^\ell}{\ell!} = e^{-\lambda}\sum_{\ell\geq 0}\frac{(\lambda s)^\ell}{\ell!} = e^{-\lambda}\, e^{\lambda s} = e^{\lambda(s-1)}.$$

The series converges for all s (infinite radius), so the equality holds on $[0,1]$ (and beyond). In particular $f(0) = e^{-\lambda} > 0$, $f(1) = 1$, $f'(s) = \lambda e^{\lambda(s-1)}$, $f''(s) = \lambda^2 e^{\lambda(s-1)} > 0$, and $f'(1) = \lambda = \mathbb{E}[\xi]$.

(b) Generating function of Z_n and mean. Write $g_n(s) = \mathbb{E}[s^{Z_n}]$. Since $Z_0 = 1$, $g_0(s) = s$. By the preliminary identity with $N = Z_n$,

$$g_{n+1}(s) = \mathbb{E}\big[s^{Z_{n+1}}\big] = \mathbb{E}\Big[\mathbb{E}\big[s^{\sum_{i=1}^{Z_n}\xi_{n,i}} \mid \mathcal{F}_n\big]\Big] = \mathbb{E}\big[f(s)^{Z_n}\big] = g_n\big(f(s)\big).$$

By induction, $g_n = \underbrace{f\circ\cdots\circ f}_{n} = f_n$ (with $g_0 = \mathrm{id}$). For the mean, we differentiate at $s=1$: $\mathbb{E}[Z_n] = g_n'(1)$. The chain rule gives $g_{n+1}'(1) = g_n'(f(1))\, f'(1) = g_n'(1)\,\lambda$ since $f(1) = 1$ and $f'(1) = \lambda$; with $g_0'(1) = 1$, we obtain

$$\mathbb{E}[Z_n] = \lambda^n \qquad (n\geq 0).$$

(This can also be obtained directly from Question 3(a) by iterating $\mathbb{E}[Z_{n+1}] = \lambda\,\mathbb{E}[Z_n]$.)

Question 2.

(a) Discussion of fixed points. The function $f(s) = e^{\lambda(s-1)}$ is continuous, strictly increasing and *strictly convex* on $[0,1]$ ($f'' > 0$), with $f(0) = e^{-\lambda} \in (0,1)$ and $f(1) = 1$. We look for the solutions of $f(s) = s$ in $[0,1]$. Set $\varphi(s) = f(s) - s$; then φ is strictly convex ($\varphi'' = f'' > 0$), with $\varphi(0) = e^{-\lambda} > 0$ and $\varphi(1) = 0$. A strictly convex function has at most two zeros; $s=1$ is always one.

- If $\lambda \leq 1$: $\varphi'(1) = f'(1) - 1 = \lambda - 1 \leq 0$. Since φ is strictly convex, φ' is strictly increasing, so $\varphi'(s) < \varphi'(1) \leq 0$ for all $s < 1$: thus φ is strictly decreasing on $[0,1)$, from the value $\varphi(0) > 0$ down to $\varphi(1) = 0$. It vanishes only at $s = 1$. The smallest (and unique) fixed point is therefore $q = 1$.

- If $\lambda > 1$: $\varphi'(1) = \lambda - 1 > 0$, so φ is strictly increasing in a left neighborhood of 1, which forces $\varphi(s) < 0$ for s slightly less than 1. Since $\varphi(0) = e^{-\lambda} > 0$ and φ is continuous, the intermediate value theorem provides a zero $q_0 \in (0,1)$. By strict convexity there are exactly two zeros, $q_0 < 1$, and the smallest fixed point is $q = q_0 \in (0,1)$.

Conclusion: $q = 1$ if $\lambda \leq 1$ and $q < 1$ if $\lambda > 1$.

Role of the critical case $\lambda = 1$. Here $\mathbb{E}[Z_n] = \lambda^n = 1$ for all n: the population has a constant mean size equal to 1. Nevertheless $f'(1) = 1$ means that the line $y = s$ is *tangent* to f at $s = 1$; since f is strictly convex, it stays *above* this tangent, so $f(s) > s$ for all $s < 1$: no interior fixed point,

$q = 1$. Extinction is almost sure. There is no contradiction: the mean $\mathbb{E}[Z_n] = 1$ is maintained by an extinction probability tending to 1 *compensated* by larger and larger values of Z_n on the (rarer and rarer) event of survival. This is typically the symptom of a failure of uniform integrability (cf. Question 3(c)).

(b) Case $\lambda = 2$. The extinction equation is $q = e^{2(q-1)}$. By (a), since $\lambda = 2 > 1$, there is a unique solution in $(0, 1)$ (the other fixed point being 1), and $q \in (0, 1)$. Let us bound it with $\varphi(s) = e^{2(s-1)} - s$. The function φ is strictly convex, $\varphi(0) = e^{-2} > 0$, $\varphi(1) = 0$ and $\varphi'(s) = 2e^{2(s-1)} - 1$ vanishes at $s_\star = 1 + \frac{1}{2}\ln\frac{1}{2} \approx 0.653$, the point where φ attains its minimum; φ is thus strictly decreasing on $[0, s_\star]$ then strictly increasing on $[s_\star, 1]$, and strictly negative between its two zeros $q \in (0, s_\star)$ and 1. We have

$$\varphi(0.2) = e^{2(0.2-1)} - 0.2 = e^{-1.6} - 0.2 \approx 0.2019 - 0.2 = +0.0019 > 0,$$

$$\varphi(0.3) = e^{2(0.3-1)} - 0.3 = e^{-1.4} - 0.3 \approx 0.2466 - 0.3 = -0.0534 < 0.$$

On the interval $[0, s_\star]$ where φ is strictly decreasing, the sign change between 0.2 and 0.3 thus locates the *small* zero: $0.2 \le q \le 0.3$ (numerically $q \approx 0.2032$).

Question 3.

(a) (W_n) is a martingale, $\mathbb{E}[W_n] = 1$. First $W_n = Z_n/\lambda^n$ is $\mathcal{F}_n$-measurable (hence adapted) and integrable since $\mathbb{E}|W_n| = \mathbb{E}[Z_n]/\lambda^n = 1 < \infty$ (Question 1(b)). For the martingale property, condition on $\mathcal{F}_n$: given $\mathcal{F}_n$, $Z_{n+1} = \sum_{i=1}^{Z_n} \xi_{n,i}$ is a sum of Z_n i.i.d. Poisson(λ) variables, independent of $\mathcal{F}_n$, whence by linearity (conditional Wald's formula, or the identity from the statement differentiated at $s = 1$)

$$\mathbb{E}\big[Z_{n+1} \mid \mathcal{F}_n\big] = Z_n \cdot \mathbb{E}[\xi] = \lambda\, Z_n \qquad \text{a.s.}$$

(also valid on $\{Z_n = 0\}$, where both sides are zero). Consequently

$$\mathbb{E}\big[W_{n+1} \mid \mathcal{F}_n\big] = \frac{1}{\lambda^{n+1}}\,\mathbb{E}\big[Z_{n+1} \mid \mathcal{F}_n\big] = \frac{\lambda Z_n}{\lambda^{n+1}} = \frac{Z_n}{\lambda^n} = W_n \quad \text{a.s.}$$

So (W_n) is a martingale; it is positive since $Z_n \ge 0$. Finally $\mathbb{E}[W_n] = \mathbb{E}[W_0] = Z_0/\lambda^0 = 1$ for all n.

(b) Almost sure convergence. (W_n) is a *positive* martingale, so $W_n \ge 0$ and $\mathbb{E}|W_n| = \mathbb{E}[W_n] = 1$: the martingale is bounded in L^1, $\sup_n \mathbb{E}|W_n| = 1 < \infty$. The martingale convergence theorem (Doob) for martingales bounded in L^1 applies: W_n converges almost surely to a random variable W_∞, finite a.s., with $W_\infty \ge 0$ (a.s. limit of nonnegative quantities) and $\mathbb{E}[W_\infty] \le \liminf_n \mathbb{E}[W_n] = 1 < \infty$ by Fatou's lemma, so W_∞ is integrable.

(c) Case $\lambda \le 1$: $W_\infty = 0$ a.s. By Question 2(a), if $\lambda \le 1$ then $q = \mathbb{P}(\text{extinction}) = 1$: almost surely there exists n_0 (random) such that $Z_n = 0$ for all $n \ge n_0$ (indeed $\{Z_n = 0\}$ is increasing in n, and the extinction event is $\bigcup_n \{Z_n = 0\}$). Hence $W_n = Z_n/\lambda^n = 0$ for all $n \ge n_0$, a.s., and passing to the limit

$$W_\infty = \lim_n W_n = 0 \qquad \text{almost surely.}$$

No L^1 convergence. We have $\mathbb{E}[W_n] = 1$ for all n (Question 3(a)), whereas $\mathbb{E}[W_\infty] = 0$. If we had $W_n \to W_\infty$ in L^1, then $\mathbb{E}[W_n] \to \mathbb{E}[W_\infty]$, i.e. $1 \to 0$, absurd. So the convergence does not hold in L^1; more precisely, since $W_\infty = 0$, $\|W_n - W_\infty\|_{L^1} = \mathbb{E}[W_n] = 1 \not\to 0$.

Comment. The martingale (W_n) converges a.s. but *not* in L^1: it is therefore *not uniformly integrable* (for a martingale, L^1 convergence $\iff$ uniform integrability $\iff$ closure by W_∞ with $\mathbb{E}[W_\infty] = \mathbb{E}[W_0]$). The mass $\mathbb{E}[W_n] = 1$ does not concentrate in the limit: it "escapes to infinity" on the survival event, of probability $\to 0$ but where Z_n becomes huge. This is exactly the phenomenon announced for the critical case in Question 2(a). $\square$

Remark (supercritical case, beyond the grading scope). For $\lambda > 1$, we have $\mathrm{Var}(\xi) = \lambda < \infty$, and one shows (by the L^2 criterion, or Kesten–Stigum under $\mathbb{E}[\xi \log^+ \xi] < \infty$, automatic here) that (W_n) is bounded in L^2, hence uniformly integrable, $\mathbb{E}[W_\infty] = 1$, and $\{W_\infty = 0\} = \{\text{extinction}\}$ up to a null set, of probability $q < 1$.

Problem V — Reflection Principle: Joint Law of the Maximum and the Hitting Time of Brownian Motion (21 pts)

Question (1).

Identity of events. Let $a > 0$. Since B has continuous paths and $B_0 = 0 < a$, on $\{M_t \geq a\}$ the process reaches the value a before t (intermediate value theorem), hence $\tau_a \leq t$ and $B_{\tau_a} = a$. Conversely, $\{\tau_a \leq t\} \subseteq \{M_t \geq a\}$. Thus

$$\{M_t \geq a\} = \{\tau_a \leq t\}.$$

Decomposition according to the terminal position. Since $\mathbb{P}(B_t = a) = 0$,

$$\mathbb{P}(M_t \geq a) = \mathbb{P}(\tau_a \leq t,\, B_t > a) + \mathbb{P}(\tau_a \leq t,\, B_t < a).$$

If $B_t > a$, by continuity $M_t \geq B_t > a$, so $\tau_a \leq t$ automatically: $\{B_t > a\} \subseteq \{\tau_a \leq t\}$, whence

$$\mathbb{P}(\tau_a \leq t,\, B_t > a) = \mathbb{P}(B_t > a).$$

Strong Markov property and symmetry. On $\{\tau_a < \infty\}$, set

$$W_s := B_{\tau_a+s} - B_{\tau_a} = B_{\tau_a+s} - a, \qquad s \geq 0.$$

By the strong Markov property at the stopping time τ_a, conditionally on $\{\tau_a < \infty\}$ the process W is a standard Brownian motion started at 0, *independent* of $\mathcal{F}_{\tau_a}$. By the symmetry of Brownian motion, $-W$ is also a Brownian motion, so W and $-W$ have the same law and remain independent of $\mathcal{F}_{\tau_a}$.

The event $A := \{\tau_a \leq t\}$ is $\mathcal{F}_{\tau_a}$-measurable, and $r := t - \tau_a$ is $\mathcal{F}_{\tau_a}$-measurable with $r \geq 0$ on A. On A we have $B_t - a = W_r$. Consider the map $\psi : (\omega, w) \mapsto \mathbf{1}_A(\omega)\, \mathbf{1}_{\{w_{r(\omega)}>0\}}$. Since W is independent of $\mathcal{F}_{\tau_a}$ and since for every $r > 0$, $W_r \sim \mathcal{N}(0, r)$ satisfies $\mathbb{P}(W_r > 0) = \mathbb{P}(W_r < 0) = \frac{1}{2}$ (and $\mathbb{P}(W_r = 0) = 0$), we obtain, by integrating first in W and then over A (where $r = t - \tau_a > 0$ a.s., the event $\{\tau_a = t\} \subseteq \{B_t = a\}$ being negligible):

$$\mathbb{P}(\tau_a \leq t,\, B_t > a) = \mathbb{P}(A,\, W_r > 0) = \mathbb{P}(A,\, W_r < 0) = \mathbb{P}(\tau_a \leq t,\, B_t < a).$$

In other words, after the first passage at a, the path is as likely to end above as below a. Combining,

$$\mathbb{P}(M_t \geq a) = 2\,\mathbb{P}(\tau_a \leq t,\, B_t > a) = 2\,\mathbb{P}(B_t > a),$$

which is the reflection principle.

Density of M_t and equality in law. For $a > 0$, since $B_t/\sqrt{t} \sim \mathcal{N}(0, 1)$,

$$\mathbb{P}(M_t \geq a) = 2\,\mathbb{P}(B_t > a) = 2\,\overline{\Phi}\big(a/\sqrt{t}\big), \qquad F_{M_t}(a) = 1 - 2\,\overline{\Phi}\big(a/\sqrt{t}\big),$$

and $M_t \geq B_0 = 0$ a.s. gives $F_{M_t}(a) = 0$ for $a < 0$. Differentiating for $a > 0$,

$$f_{M_t}(a) = \frac{2}{\sqrt{t}}\,\varphi\big(a/\sqrt{t}\big) = \sqrt{\tfrac{2}{\pi t}}\; e^{-a^2/(2t)}, \qquad a > 0,$$

zero for $a < 0$. Finally, by the symmetry of B_t, for $a > 0$,

$$\mathbb{P}(|B_t| \geq a) = \mathbb{P}(B_t \geq a) + \mathbb{P}(B_t \leq -a) = 2\,\mathbb{P}(B_t > a) = \mathbb{P}(M_t \geq a),$$

and since $M_t, |B_t| \geq 0$, these two nonnegative variables have the same survival function: $M_t \overset{d}{=} |B_t|$.

Question (2).

Let $a > 0$ and $b \le a$. We again take $W_s = B_{\tau_a+s} - a$ (Brownian motion started at 0, independent of $\mathcal{F}_{\tau_a}$) and $r = t - \tau_a$, so that on $A = \{\tau_a \le t\} = \{M_t \ge a\}$ we have $B_t = a + W_r$. Since $b - a \le 0$ and, conditionally on $\mathcal{F}_{\tau_a}$, $W_r \overset{d}{=} -W_r$,

$$\mathbb{P}\big(M_t \ge a,\ B_t \le b\big) = \mathbb{P}\big(A,\ W_r \le b - a\big) = \mathbb{P}\big(A,\ W_r \ge a - b\big).$$

Now if $W_r \ge a - b \ge 0$, then $B_t = a + W_r \ge 2a - b \ge a > 0$, which forces $M_t \ge a$ and hence A; the constraint A is thus redundant:

$$\mathbb{P}\big(M_t \ge a,\ B_t \le b\big) = \mathbb{P}\big(B_t \ge 2a - b\big).$$

(Interpretation: reflecting the path after τ_a sends an endpoint at b onto its mirror image $2a - b$ with respect to the level a, preserving the law.)

Joint density. Write $p_t(x) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/2t}$, so that $p_t'(x) = -\frac{x}{t} p_t(x)$, and

$$G(a, b) := \mathbb{P}(M_t \ge a,\ B_t \le b) = \int_{2a-b}^{\infty} p_t(x)\, dx.$$

The function G is increasing in b and decreasing in a; the joint density is $f_{(M_t,B_t)}(a, b) = -\partial^2 G/\partial a\, \partial b$. We have

$$\frac{\partial G}{\partial b} = p_t(2a - b), \qquad \frac{\partial^2 G}{\partial a\, \partial b} = 2\, p_t'(2a - b) = -\frac{2(2a - b)}{t}\, p_t(2a - b),$$

whence, on $\{a \ge 0,\ \ b \le a\}$,

$$f_{(M_t,B_t)}(a, b) = \frac{2(2a - b)}{t}\, p_t(2a - b) = \frac{2(2a - b)}{\sqrt{2\pi}\, t^{3/2}} \exp\Big(- \frac{(2a - b)^2}{2t} \Big),$$

and 0 elsewhere. On this domain $2a - b \ge a \ge 0$, so the density is positive.

Verification. For fixed $a > 0$, with $u = 2a - b$ (u ranges over $[a, \infty)$ as $b \le a$):

$$\int_{-\infty}^{a} f_{(M_t,B_t)}(a, b)\, db = \int_{a}^{\infty} \frac{2u}{t}\, p_t(u)\, du = 2\big[- p_t(u)\big]_a^\infty = 2\, p_t(a) = \sqrt{\tfrac{2}{\pi t}}\, e^{-a^2/2t} = f_{M_t}(a),$$

in agreement with question (1).

Question (3).

Identity of events. Established in (1): for $a > 0$, $\{M_t \ge a\} = \{\tau_a \le t\}$ (continuity $\Rightarrow$ intermediate value $\Rightarrow$ $B_{\tau_a} = a$ and $\tau_a \le t$, and conversely).

Density of τ_a. Hence for $s > 0$ (writing s for the time horizon),

$$F_{\tau_a}(s) = \mathbb{P}(\tau_a \le s) = \mathbb{P}(M_s \ge a) = 2\, \overline{\Phi}\big(a/\sqrt{s}\big) = 2 \int_{a/\sqrt{s}}^{\infty} \varphi(y)\, dy.$$

Differentiating in s, with $\frac{d}{ds}\big(a\, s^{-1/2}\big) = -\frac{1}{2} a\, s^{-3/2}$,

$$f_{\tau_a}(s) = -2\, \varphi\big(a/\sqrt{s}\big) \cdot \Big(- \tfrac{1}{2} a\, s^{-3/2} \Big) = a\, s^{-3/2}\, \varphi\big(a/\sqrt{s}\big) = \frac{a}{\sqrt{2\pi}\, s^{3/2}} \exp\Big(- \frac{a^2}{2s} \Big), \qquad s > 0.$$

(Stable law of index 1/2, the so-called Lévy law.)

Almost sure finiteness.

$$\mathbb{P}(\tau_a < \infty) = \lim_{s \to \infty} F_{\tau_a}(s) = \lim_{s \to \infty} 2\, \overline{\Phi}\big(a/\sqrt{s}\big) = 2\, \overline{\Phi}(0) = 1.$$

Brownian motion therefore reaches every level $a > 0$ almost surely (by symmetry, every real level).

Infinite expectation. We have $f_{\tau_a}(s) = \frac{a}{\sqrt{2\pi}}\, s^{-3/2} e^{-a^2/2s}$, and for $s \ge 1$, $e^{-a^2/2s} \ge e^{-a^2/2} > 0$. Hence

$$\mathbb{E}[\tau_a] = \int_0^\infty s\, f_{\tau_a}(s)\, ds \ \ge\ \frac{a\, e^{-a^2/2}}{\sqrt{2\pi}} \int_1^\infty s \cdot s^{-3/2}\, ds = \frac{a\, e^{-a^2/2}}{\sqrt{2\pi}} \int_1^\infty s^{-1/2}\, ds = +\infty.$$

Thus $\tau_a < \infty$ a.s. but $\mathbb{E}[\tau_a] = +\infty$: this is the signature of the heavy tail $\mathbb{P}(\tau_a > s) \sim \sqrt{2/\pi}\, (a/\sqrt{s})$ as $s \to \infty$.